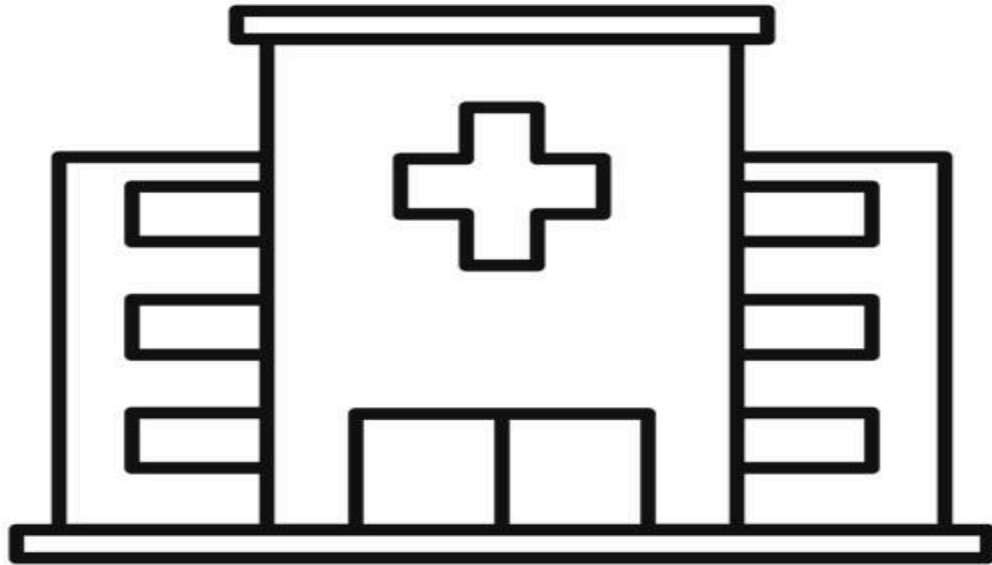


Hospital Management System

Database Management Systems Project



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Introduction

A new hospital requires assistance in creating and maintaining a database to manage its local community's patients. This hospital does not perform surgeries, but instead focuses on diagnosing patients and prescribing them with medication when necessary. The hospital also has staff that needs to be allocated to their respective functions within this new database, eg. receptionist to the appointment system. This database also needs to be consistently updated, such as when appointments are made and finished or when patient health records need to be changed.

To be included in our database are tables: staff, receptionist, nurse, doctor, appointment, visit, patient, treatment history, treatment type, prescription, prescription history, pharmacy, payment, and insurance provider.

Executive Summary

A new hospital created a database to manage its local community's patients. This hospital focuses on diagnosing patients and prescribing them with medication. In creating a database for this hospital, we have successfully streamlined their operations and ensured its efficacy. Each member of our team contributed equally, allowing for us to pool our ideas together to create an efficient system for this hospital.

The first step in creating the database involved learning about a hospital and its database requirements. We had to evaluate which tables a hospital would need, and its associated attributes. Then, we proceeded to create the Entity Relationship Model (ERM). This shows the tables and their associated relationships with one another, allowing for a visual view of how the hospital processes work. It also highlights whether relationships are one to many or many to many. We also created relationship sentences for each relationship between tables.

Then, we put the ERM in RDM format, which allows us to easily show each tables' primary and foreign keys.

The next step in our process was to normalize our tables and relationships to BCNF. This helps in reducing anomalies, which maintains the integrity in our data, and ensures that the hospital database we created.

Continuing forward, we had to create a database using Structured Query Language (SQL). Our first step was to create all the tables using the CREATE TABLE function, and adding all the attributes with their respective tables. For the attributes, we made sure to specify what type of attribute it was, and whether or not if it was NULL or NOT NULL, depending on the attribute itself.

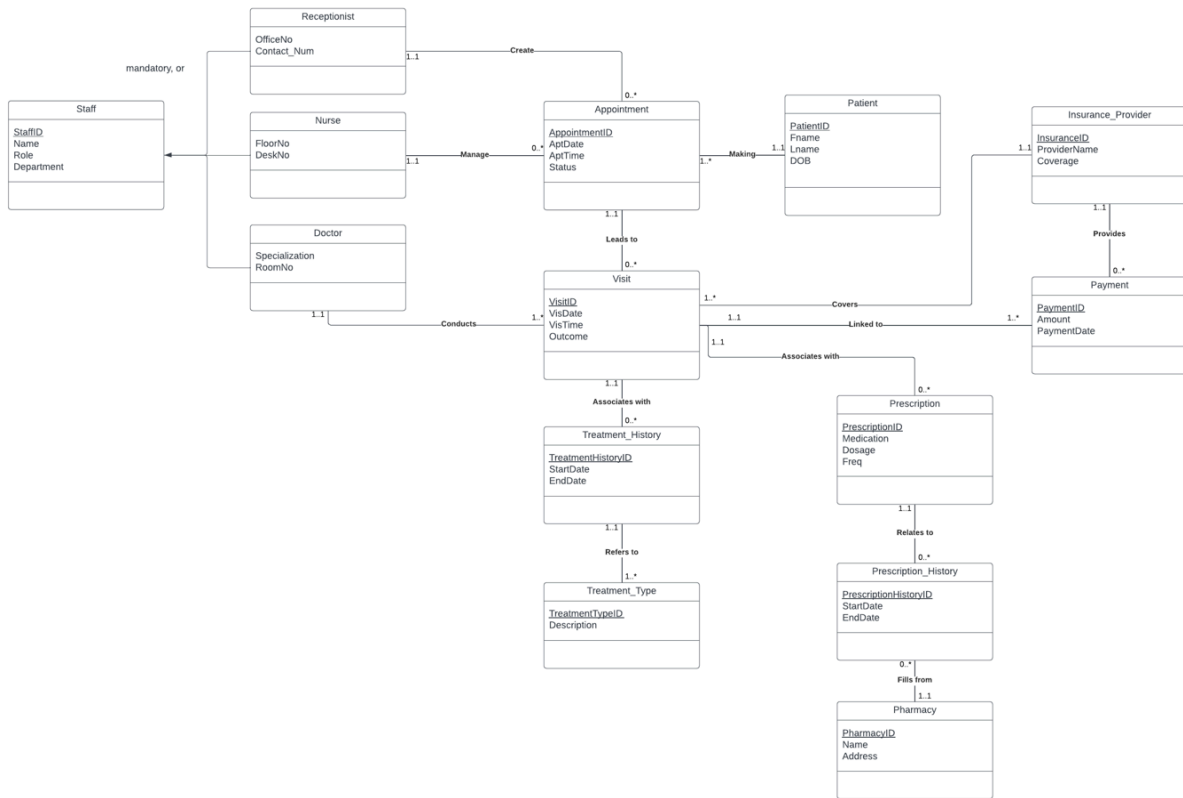
After the table creation code, our final step was to insert data using the INSERT INTO function. Each table had around five rows of data. This was also done in preparation for our final step, scenario creation and code implementation.

To test the efficacy of our database, we created five scenarios that would be useful in a hospital environment. They were as follows:

1. The hospital is looking for patients that may have payments due over 30 days; as such, they will look for patient ID, amount, payment date, where payment date exceeds visit date by 30 days.
2. There has been a surplus in the hospital budget, and the Head of Medicine is trying to determine the headcount of nurses currently employed. They need a query to list the names of the nurses, along with their respective departments and floor numbers.
3. To see availability, the hospital is looking for closest appointment dates with patient and nurse.
4. Average prescription length using prescription_history table to be used for internal analysis.
5. Insurance providers that give full coverage to be used for internal analysis.

We then created the code for these scenarios and documented our results.

ERD Model



Relationship Sentences

Superclass: Staff; Subclass: Receptionist, Doctor, Nurse; Mandatory, or

One **receptionist** may create zero to many **appointments**. 0..*

One **appointment** can be created by one and only one **receptionist**. 1..1

One **nurse** may manage zero to many **appointments**. 0..*

One **appointment** may be managed by one and only one nurse. 1..1

One **appointment** may lead to zero to many **visits**. 0..*

One **visit** may be created by one and only one **appointment**. 1..1

One **appointment** may be made by one and only one **patient**. 1..1

One **patient** may make one to many **appointments**. 1..*

One **insurance provider** provides zero to many **payments**. 0..*

One **insurance provider** can cover one or more **visits**. 1..*

One **visit** can be covered by only one **insurance provider**. 1..1

One **visit** can be linked to one or more **payment options**. 1..*

One **payment** occurrence can be for only one particular **visit**. 1..1

One **payment** is provided by one and only one **insurance provider**. 1..1

One **doctor** may conduct one to many **visits**. 1..*

One **visit** may be conducted by one and only one **doctor**. 1..1

One **visit** is associated with zero to many **treatment histories**. 0..*

One **treatment history** is associated with one and only one **visit**. 1..1
 One **treatment history** is associated with one to many **treatment types**. 1..*
 One **treatment type** is associated with one **treatment**. 1..1
 One **visit** is associated with zero to many **prescriptions**. 0..*
 One **prescription** is associated with one and only one **visit**. 1..1
 One **prescription** relates to zero to many **prescription histories**. 0..*
 One **prescription history** relates to one and only one **prescription**. 1..1
 One **prescription history** is filled by one and only one **pharmacy**. 1..1
 One **pharmacy** may fill zero to many **prescription histories**. 0..*

RDM

Staff - (StaffID, Name, Role, Department)
Receptionist - (StaffID, OfficeNo, Contact_Num)
Nurse - (StaffID, FloorNo, DeskNo)
Doctor - (StaffID, Specialization, RoomNo)
Patient - (PatientID, Fname, Lname, DOB)
Appointment - (AppointmentID, AptDate, AptTime, Status, PatientID(FK), StaffID(FK))
Visit - (VisitID, VisDate, ViTime, Outcome, StaffID(FK), InsuranceID (FK), AppointmentID(FK))
Insurance _Provider - (InsuranceID, ProviderName, Coverage)
Payment - (PaymentID, Amount, PaymentDate, InsuranceID(FK), VisitID(FK))
Treatment _History - (TreatmentHistoryID, StartDate, EndDate, VisitID (FK))
Treatment _Type - (TreatmentTypeID, Description, TreatmentHistoryID(FK))
Prescription - (PrescriptionID, Medication, Dosage, Freq, VisitID(FK))
Prescription _History - (PrescriptionHistoryID, StartDate, EndDate, PrescriptionID(FK), PharmacyID(FK))
Pharmacy - (PharmacyID, Name, Address)

Normalization

Staff - StaffID Primary Key so 1NF
 StaffID -> Name, Role, Department
 Already 2NF, 3NF and BCNF

Receptionist - StaffID Primary Key so 1NF
 StaffID -> OfficeNo, Contact_Num
 Already 2NF, 3NF and BCNF

Nurse - StaffID Primary Key so 1NF
 StaffID -> FloorNo, DeskNo
 Already 2NF, 3NF and BCNF

Doctor - StaffID Primary Key so 1NF

StaffID -> RoomNo, Specialization

Already 2NF, 3NF and BCNF

Insurance_Provider - InsuranceID -> ProviderName, Coverage

Payment - PaymentID -> Amount, PaymentDate, VisitID

No partial or transitive dependency for Insurance provider and Payment

Already in BCNF form

Visit - (VisitID, VisDate, ViTime, Outcome, StaffID(FK), InsuranceID (FK), AppointmentID(FK))

VisitID -> VisDATE, ViTime, Outcome, StaffID, InsuranceID, AppointmentID

Prescription - (PrescriptionID, Medication, Dosage, Freq, VisitID(FK))

PrescriptionID -> Medication, Dosage, Freq, VisitID

Already 2NF, 3NF and BCNF

Treatment_History - (TreatmentHistoryID, StartDate, EndDate, VisitID (FK))

TreatmentHistoryID -> StartDate, EndDate, VisitID

VisitID -> TreatmentHistoryID

Treatment_Type - (TreatmentTypeID, Description, TreatmentHistoryID(FK))

TreatmentTypeID -> Description, TreatmentHistoryID

Appointment - (Appointment ID, AptDate, AptTime, Status, PatientID(FK), StaffID(FK))

AppointmentID -> AptDate, AptTime, Status, PatientID, StaffID

Patient - (PatientID, Fname, Lname, DOB)

PatientID -> Fname, Lname, DOB

Prescription_History - PrescriptionHistoryID -> StartDate, EndDate, PrescriptionID(fk), PharmacyID(fk)

Pharmacy - PharmacyID -> Name, Address

All tables were normalized to BCNF.

SQL DDL

Staff

```
CREATE TABLE Staff(  
StaffID NUMBER NOT NULL,  
Name VARCHAR(120),  
Role VARCHAR(120),  
Department VARCHAR(120),  
OfficeNo NUMBER,  
Contact_Num VARCHAR(120),  
FloorNo NUMBER,  
DeskNo VARCHAR(120),  
Specialization VARCHAR(120),  
RoomNo NUMBER,
```

```
CONSTRAINT pk_staff PRIMARY KEY (StaffID))
```

```
INSERT INTO Staff  
VALUES (1000001,'Emily Parker','Nurse',' Pediatric', 202,'(123)  
456-7890',2,'2-210','Nursing',2210);  
INSERT INTO Staff  
VALUES(1000002,'Samuel Rodriguez','Doctor','ICU',101,'(987) 654-3210',  
1,'1-100','Critical Care',1100);  
INSERT INTO Staff  
VALUES(1000003,'Jessica Thompson','Doctor','Pediatric',201,'(234)  
567-8901',2,'2-200','Pediatrics',2200);  
INSERT INTO Staff  
VALUES(1000004,'Kevin Nguyen','Nurse','ICU',102,'(876) 543-2109',1,'  
1-110','Nursing',1110);  
INSERT INTO Staff  
VALUES (1000013,'Isabella Harris','Nurse','Pediatric',202,'(876) 543-2109',  
2,'2-211','Nursing',2210);
```

	StaffID	Name	Role	Department	OfficeNo	Contact_Num	FloorNo	DeskNo	Specialization	RoomNo
+	1000001	Emily Parker	Nurse	Pediatric	202	(123) 456-7890	2	2-210	Nursing	2210
+	1000002	Samuel Rodriguez	Doctor	ICU	101	(987) 654-3210	1	1-100	Critical Care	1100
+	1000003	Jessica Thompson	Doctor	Pediatric	201	(234) 567-8901	2	2-200	Pediatrics	2200
+	1000004	Kevin Nguyen	Nurse	ICU	102	(876) 543-2109	1	1-110	Nursing	1110
+	1000013	Isabella Harris	Nurse	Pediatric	202	(876) 543-2109	2	2-211	Nursing	2210
✱										

Patient

```
CREATE TABLE Patient(  
PatientID NUMBER NOT NULL,  
Fname VARCHAR(255),
```

Lname VARCHAR(255),
 DOB DATE,
 CONSTRAINT pk_Patient PRIMARY KEY (PatientID)) ;

INSERT INTO Patient
 VALUES (50001, 'Emily', 'Smith', '9/23/2009');
 INSERT INTO Patient
 VALUES (50005, 'Noah', 'Johnson', '5/10/1995');
 INSERT INTO Patient
 VALUES (50010, 'Sophia', 'Williams', '7/6/1987');
 INSERT INTO Patient
 VALUES (50014, 'Liam', 'Jones', '11/18/1982');
 INSERT INTO Patient
 VALUES (50019, 'Isabella', 'Brown', '2/14/1980');

	PatientID	Fname	Lname	DOB
+	50001	Emily	Smith	9/23/2009
+	50005	Noah	Johnson	5/10/1995
+	50010	Sophia	Williams	7/6/1987
+	50014	Liam	Jones	11/18/1982
+	50019	Isabella	Brown	2/14/1980
*				

Appointment

CREATE TABLE Appointment(
 AppointmentID NUMBER NOT NULL,
 AptDate DATE,
 AptTime TIME,
 Status VARCHAR(255),
 StaffID NUMBER NOT NULL,
 PatientID NUMBER NOT NULL,

CONSTRAINT pk_AppointmentID PRIMARY KEY (AppointmentID),
 CONSTRAINT fk_Appointment1 FOREIGN KEY (StaffID) REFERENCES Staff(StaffID),
 CONSTRAINT fk_Appointment2 FOREIGN KEY (PatientID) REFERENCES
 Patient(PatientID));

INSERT INTO Appointment
 VALUES (503001, '5/12/2024', '8:00 A.M', 'Complete', 1000004, 50001);


```

INSERT INTO Appointment
VALUES (503002,'5/12/2024','10:00 A.M','Complete',1000004,50005);
INSERT INTO Appointment
VALUES (503003,'5/12/2024','1:00 P.M','Complete',1000004,50010);
INSERT INTO Appointment
VALUES (503004,'5/13/2024','8:00 A.M','Returning',1000004,50014);
INSERT INTO Appointment
VALUES (503005,'5/13/2024','9:00 A.M','Complete',1000013,50019);

```

	Appointment ▾	AptDate ▾	AptTime ▾	Status ▾	StaffID ▾	PatientID ▾
+	503001	5/12/2024	8:00 A.M	Complete	1000004	50001
+	503002	5/12/2024	10:00 A.M	Complete	1000004	50005
+	503003	5/12/2024	1:00 P.M	Complete	1000004	50010
+	503004	5/13/2024	8:00 A.M	Returning	1000004	50014
+	503005	5/13/2024	9:00 A.M	Complete	1000013	50019

Insurance Provider

```

CREATE TABLE Insurance_Provider
(
InsuranceID NUMBER NOT NULL,
ProviderName VARCHAR(255),
Coverage VARCHAR(255) ,
CONSTRAINT Pk_Insurance_Provider PRIMARY KEY (InsuranceID)
);

```

```

INSERT INTO Insurance_Provider
VALUES (164207531, 'United Healthcare', 'Full');
INSERT INTO Insurance_Provider
VALUES (325879604, 'Aptos', 'Full');
INSERT INTO Insurance_Provider
VALUES (478613295, 'Aptos', 'Half');
INSERT INTO Insurance_Provider
VALUES (590812036, 'United Healthcare', 'Full' );
INSERT INTO Insurance_Provider
VALUES (783405920, 'United Healthcare' , 'Full' );

```

	InsuranceID ▼	ProviderName ▼	Coverage ▼
+	164207531	United Healthcare	Full
+	325879604	Aptos	Full
+	478613295	Aptos	Half
+	590812036	United Healthcare	Full
+	783405920	United Healthcare	Full

Visit

```
CREATE TABLE Visit(
    VisitID NUMBER NOT NULL,
    Date_of_Visit DATE,
    Time_of_Visit VARCHAR(20),
    Outcome VARCHAR(20),
    StaffID NUMBER NOT NULL,
    InsuranceID NUMBER NOT NULL,
    AppointmentID NUMBER NOT NULL,

    CONSTRAINT pk_Visit PRIMARY KEY (VisitID),
    CONSTRAINT fk_Visit1 FOREIGN KEY (StaffID) REFERENCES Staff(StaffID),
    CONSTRAINT fk_Visit2 FOREIGN KEY (InsuranceID) REFERENCES Insurance_Provider
    (InsuranceID),
    CONSTRAINT fk_Visit3 FOREIGN KEY (AppointmentID) REFERENCES Appointment
    (AppointmentID)
);
```

```
INSERT INTO Visit
VALUES (7000001, '5/12/2024', '8:00 A.M', 'Complete',1000004,590812036,503001);
INSERT INTO Visit
VALUES (7000002, '5/12/2024', '10:00 A.M' , 'Complete',1000004,783405920,503002);
INSERT INTO Visit
VALUES (7000003, '5/12/2024', '1:00 P.M', 'Complete',1000004,164207531,503003);
INSERT INTO Visit
VALUES (7000004, '5/13/2024', '8:00 A.M', 'Returning',1000004,325879604,503004);
INSERT INTO VisitVALUES (7000005, '5/13/2024' , '9:00 A.M',
'Complete',1000013,478613295,503005);
```

	VisitID	Date_of_Visi	Time_of_Visi	Outcome	StaffID	InsuranceID	Appointment
+	7000001	5/12/2024	8:00 A.M	Complete	1000004	590812036	503001
+	7000002	5/12/2024	10:00 A.M	Complete	1000004	783405920	503002
+	7000003	5/12/2024	1:00 P.M	Complete	1000004	164207531	503003
+	7000004	5/13/2024	8:00 A.M	Returning	1000004	325879604	503004
+	7000005	5/13/2024	9:00 A.M	Complete	1000013	478613295	503005

Treatment History

CREATE TABLE Treatment_History

(

TreatmentHistoryID NUMBER NOT NULL,

StartDate DATE,

EndDate DATE,

VisitID NUMBER NOT NULL,

CONSTRAINT pk_Treatment_History PRIMARY KEY (TreatmentHistoryID),

CONSTRAINT fk_Visit FOREIGN KEY (VisitID) REFERENCES Visit (VisitID)

);

INSERT INTO Treatment_History

VALUES (1234, '4/12/2024', '5/12/2024', 7000001)

INSERT INTO Treatment_History

VALUES (3456, '12/4/2018', '5/13/2024', 7000004)

INSERT INTO Treatment_History

VALUES (5678, '5/1/2024', '5/12/2024', 7000002)

INSERT INTO Treatment_History

VALUES (7890, '4/30/2024', '5/17/2024', 7000005)

INSERT INTO Treatment_History

VALUES (9012, '5/10/2024', '5/12/2024', 7000003)

	TreatmentHistoryID	StartDate	EndDate	VisitID
+	1234	4/12/2024	5/12/2024	7000001
+	3456	12/4/2018	5/13/2024	7000004
+	5678	5/1/2024	5/12/2024	7000002
+	7890	4/30/2024	5/17/2024	7000005
+	9012	5/10/2024	5/12/2024	7000003

Treatment Type

CREATE TABLE Treatment_Type

(

TreatmentTypeID NUMBER NOT NULL,

Description VARCHAR (100),

TreatmentHistoryID NUMBER NOT NULL,

```

CONSTRAINT pk_Treatment_Type PRIMARY KEY (TreatmentTypeID),
CONSTRAINT fk_Treatment_History FOREIGN KEY (TreatmentHistoryID) REFERENCES
Treatment_History (TreatmentHistoryID)
);

```

```

INSERT INTO Treatment_Type
VALUES (101, 'Delayed Speech and Motor Skills Disorder', 1234);
INSERT INTO Treatment_Type
VALUES (202, 'Acute Respiratory Distress Syndrome', 5678);
INSERT INTO Treatment_Type
VALUES (303, 'Pediatric Seasonal Influenza', 9012);
INSERT INTO Treatment_Type
VALUES (404, 'Chronic Kidney Failure, Stage 5', 3456);
INSERT INTO Treatment_Type
VALUES (505, 'Post Cardiac Arrest Syndrome', 7890);

```

TreatmentTy	Description	TreatmentHi
101	Delayed Speech and Motor Skills Disorder	1234
202	Acute Respiratory Distress Syndrome	5678
303	Pediatric Seasonal Influenza	9012
404	Chronic Kidney Failure, Stage 5	3456
505	Post Cardiac Arrest Syndrome	7890

Payment

```

CREATE TABLE Payment (
PaymentID NUMBER NOT NULL,
Amount DECIMAL(10, 2),
PaymentDate DATE,
VisitID NUMBER,
CONSTRAINT Payment_PaymentID PRIMARY KEY (PaymentID),
CONSTRAINT Payment_FK FOREIGN KEY (VisitID) REFERENCES Visit(VisitID));

```

```

INSERT INTO Payment
VALUES (123456789012, 25.67, '5/10/2024', 7000001)
INSERT INTO Payment
VALUES (123456789034, 39.99, '5/11/2024', 7000002)
INSERT INTO Payment
VALUES (123456789056, 17.85, '5/12/2024', 7000003)
INSERT INTO Payment
VALUES (123456789078, 62.50, '5/13/2024', 7000004)

```

```
INSERT INTO Payment
VALUES (87654321098, 45.75, '5/11/2024', 7000005)
```

PaymentID	Amount	PaymentDate	VisitID
87654321098	45.75	5/11/2024	7000005
123456789012	25.67	5/10/2024	7000001
123456789034	39.99	5/11/2024	7000002
123456789056	17.85	5/12/2024	7000003
123456789078	62.5	5/13/2024	7000004

Prescription

```
CREATE TABLE Prescription (
PrescriptionID NUMBER NOT NULL,
Medication VARCHAR(20) NOT NULL,
Dosage_mg NUMBER,
Freq VARCHAR(30),
VisitID NUMBER NOT NULL,
```

```
CONSTRAINT pk_Prescription PRIMARY KEY (PrescriptionID),
CONSTRAINT fk_Prescription1 FOREIGN KEY (VisitID) REFERENCES Visit(VisitID)
);
```

```
INSERT INTO Prescription
VALUES (1234123456, "Acetaminophen", 500, "2 tablets per 6 hrs", 7000001);
INSERT INTO Prescription
VALUES (5647234567, "Metformin", 1000, "1 tablet twice daily", 7000002);
INSERT INTO Prescription
VALUES (8905345678, "Simvastatin", 20, "1 tablet daily", 7000003);
INSERT INTO Prescription
VALUES (12933456789, "Omeprazole", 40, "1 capsule daily", 7000004);
INSERT INTO Prescription
VALUES (16769067900, "Acetaminophen", 500, "2 tablets per 6 hrs", 7000005);
```

PrescriptionID	Medication	Dosage_mg	Freq	VisitID
1234123456	Acetaminophen	500	2 tablets per 6 h	7000001
5647234567	Metformin	1000	1 tablet twice d	7000002
8905345678	Simvastatin	20	1 tablet daily	7000003
12933456789	Omeprazole	40	1 capsule daily	7000004
16769067900	Acetaminophen	500	2 tablets per 6 h	7000005

Prescription_History

```
CREATE TABLE Prescription_History(  
PrescriptionHistoryID NUMBER NOT NULL,  
StartDate DATE,  
EndDate DATE,  
PrescriptionID NUMBER,  
PharmacyID NUMBER,  
CONSTRAINT PK_Prescription_History PRIMARY KEY (PrescriptionHistoryID),  
CONSTRAINT FK1_Prescription_History FOREIGN KEY (PrescriptionID) REFERENCES  
Prescription(PrescriptionID),  
CONSTRAINT FK2_Prescription_History FOREIGN KEY (PharmacyID) REFERENCES  
Pharmacy(PharmacyID)  
);
```

```
INSERT INTO Prescription_History  
VALUES (301051001, "5/12/2024", "5/17/2024", 1234123456, 210134);  
INSERT INTO Prescription_History  
VALUES (301051002, "5/12/2024", "7/21/2025", 5647234567, 210256);  
INSERT INTO Prescription_History  
VALUES (301051003, "5/12/2024", "2/1/2025", 8905345678, 210378);  
INSERT INTO Prescription_History  
VALUES (301051004, "5/13/2024", "7/21/2024", 12933456789, 210500);  
INSERT INTO Prescription_History  
VALUES (301051005, "5/13/2024", "5/17/2024", 16769067900, 210622);
```

PrescriptionHistoryID	StartDate	EndDate	PrescriptionID	PharmacyID
301051001	5/12/2024	5/17/2024	1234123456	210134
301051002	5/12/2024	7/21/2025	5647234567	210256
301051003	5/12/2024	2/1/2025	8905345678	210378
301051004	5/13/2024	7/21/2024	12933456789	210500
301051005	5/13/2024	5/17/2024	16769067900	210622

Pharmacy

```
CREATE TABLE Pharmacy(  
PharmacyID number NOT NULL,  
Name varchar(120),  
Address varchar(150),  
CONSTRAINT PK_Pharmacy PRIMARY KEY (PharmacyID)  
)
```

```
INSERT INTO Pharmacy
```

```
VALUES (210134, "Springfield Pharmacy", "123 Main Street, Springfield, New York, 12345");
INSERT INTO Pharmacy
VALUES (210256, "HealthLink Pharmacy", "456 Elm Street, Springfield, New York, 12346");
INSERT INTO Pharmacy
VALUES (210378, "QuickCare Pharmacy", "789 Oak Avenue, Springfield, New York, 12347");
INSERT INTO Pharmacy
VALUES (210500, "VitalRx Pharmacy", "321 Maple Lane, Springfield, New York, 12348");
INSERT INTO Pharmacy
VALUES (210622, "Springfield Family Pharmacy", "567 Pine Street, Springfield, New York, 12349");
```

	PharmacyID	Name	Address
+	210134	Springfield Phar	123 Main Street, Springfield, New York, 12345
+	210256	HealthLink Phar	456 Elm Street, Springfield, New York, 12346
+	210378	QuickCare Phar	789 Oak Avenue, Springfield, New York, 12347
+	210500	VitalRx Pharmac	321 Maple Lane, Springfield, New York, 12348
+	210622	Springfield Fami	567 Pine Street, Springfield, New York, 12349

Alter Table

```
ALTER TABLE Prescription
ALTER COLUMN Freq VARCHAR(120);
```

	PrescriptionID	Medication	Dosage_mg	Freq	VisitID
+	1234123456	Acetaminophen	500	2 tablets per 6 hrs	7000001
+	5647234567	Metformin	1000	1 tablet twice daily	7000002
+	8905345678	Simvastatin	20	1 tablet daily	7000003
+	12933456789	Omeprazole	40	1 capsule daily	7000004
+	16769067900	Acetaminophen	500	2 tablets per 6 hrs	7000005

Scenarios

Query: The hospital is looking for patients that may have payments due over 30 days; as such, they will look for patient ID, amount, payment date, where payment date exceeds visit date by 30 days.

```
SELECT Appointment.PatientID, Payment.Amount, Payment.PaymentDate
FROM ((Payment
INNER JOIN Visit ON Payment.VisitID = Visit.VisitID)
INNER JOIN Appointment ON Visit.AppointmentID = Appointment.AppointmentID)
WHERE DateDiff("d", Visit.Date_of_Visit, Payment.PaymentDate) >= 30;
```

	PatientID ▾	Amount ▾	PaymentDate ▾
*			

(Note: None! All payments have been made within 30 days)

Query: There has been a surplus in the hospital budget, and the Head of Medicine is trying to determine the headcount of nurses currently employed. They need a query to list the names of the nurses, along with their respective departments and floor numbers.

```
SELECT
Name, Department, FloorNo
FROM
Staff
WHERE
Role LIKE '*Nurse*';
```

	Name ▾	Department ▾	FloorNo ▾
	Emily Parker	Pediatric	2
	Kevin Nguyen	ICU	1
	Isabella Harris	Pediatric	2

Query: To see availability, the hospital is looking for closest appointment dates with patient and nurse.

```
SELECT a.AptDate, p.Fname AS Patient_First_Name, p.Lname AS Patient_Last_Name, s.Name
AS Nurse
FROM Appointment a, Patient p, Staff s
WHERE a.StaffID = s.StaffID AND a.PatientID = p.PatientID
ORDER BY a.AptDate ASC
```

	AptDate ▾	Patient_Firs ▾	Patient_Last ▾	Nurse ▾
	5/12/2024	Sophia	Williams	Kevin Nguyen
	5/12/2024	Noah	Johnson	Kevin Nguyen
	5/12/2024	Emily	Smith	Kevin Nguyen
	5/13/2024	Isabella	Brown	Isabella Harris
	5/13/2024	Liam	Jones	Kevin Nguyen

Query: Average prescription length using prescription_history table to be used for internal analysis.


```
SELECT Avg(DateDiff("d", [StartDate], [EndDate])) AS Average_Prescription_Length_In_Days
FROM Prescription_History;
```

Average_Pre!
155.6

Query: Insurance providers that give full coverage to be used for internal analysis.

```
SELECT ProviderName
FROM Insurance_Provider
WHERE Coverage = "Full";
```

ProviderName
United Healthcare
Aptos
United Healthcare
United Healthcare
*

Conclusion

In this project, we learned about the various components of database modeling and went through each step meticulously to ensure our database functioned properly and accurately. It was interesting to learn about these components and how they fit in with one another to create an effective and efficient database. In working with the database, we also learned about how a typical hospital functions, and how a database like this can streamline a lot of its processes. Through our scenario creation and implementation, we learned how specific scenarios can be used for data analysis for the hospital itself, such as looking for particular payments that are overdue by a month.

The entire project was a group effort, with each of us contributing our best to the creation of the database. We kept each other updated on our progress on our respective tasks, and made sure that each of us was held accountable for the work we submitted. We hope that the database created will provide high value for the hospital in its current and future organizational processes.